

Student number: _____

2022 Trial Higher School Certificate Examination

Mathematics Advanced

General Instructions

- Reading time - 10 minutes
- Working time - 3 hours
- Write using black or blue pen
- Calculators approved by NESA may be used
- A reference sheet is provided at the back of this paper
- For questions in Section II, show relevant mathematical reasoning and/or calculations

Total marks – 100 Section I - 10 marks (pages 2- 6)

- Attempt Questions 1-10
- Allow about 15 minutes for this section

Section II - 90 marks (pages 8 - 34)

- Attempt Questions 11- 35
- Allow about 2 hours and 45 minutes for this section

Care has been taken to ensure that this paper is free of errors and that it mirrors the format and style of past HSC papers. The questions have been adapted from various sources, in an attempt to provide students with exposure to a broad range of questions. However, there is no guarantee whatsoever that the HSC examination will have similar content, style or format. This paper is intended only as a trial for the HSC examination or as revision leading to the examination.

Section I

10 marks

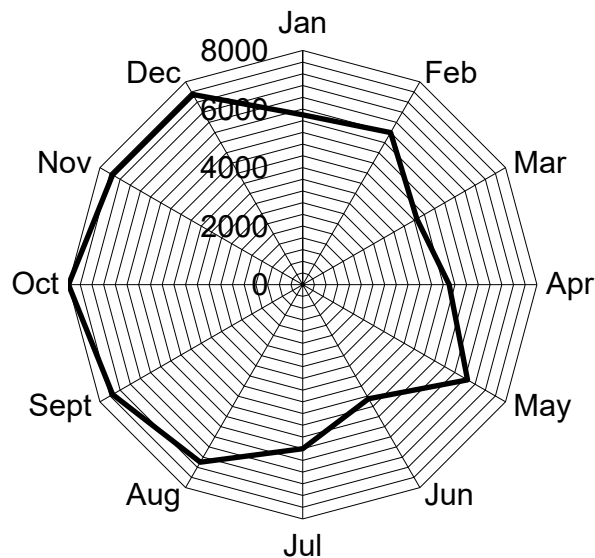
Attempt Questions 1 – 10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1- 10

Question 1

The following radar chart shows the monthly sales figures for a local bookshop.



Which two months have the lowest sales?

- (A) Nov/Dec
- (B) Oct/Sep
- (C) Jul/Mar
- (D) Jun/Mar

Question 2

John's marks, the class mean and standard deviation in three tests are given in the table.

	Mathematics	English	Science
John's mark	59	60	70
Class mean	55	50	72
Class standard deviation	4	5	2

Which of the following would be a correct conclusion.

- (A) He did better in Science compared to English
- (B) He did equally well in English and Science
- (C) He did better in Mathematics compared to English
- (D) He did better in English compared to Mathematics

Question 3

A special deck of cards consist of 3 red and 3 black cards. Two cards are selected at random without replacement. The probability of choosing cards the same colour is?

- (A) $\frac{1}{50}$
- (B) $\frac{1}{25}$
- (C) $\frac{1}{5}$
- (D) $\frac{2}{5}$

Question 4

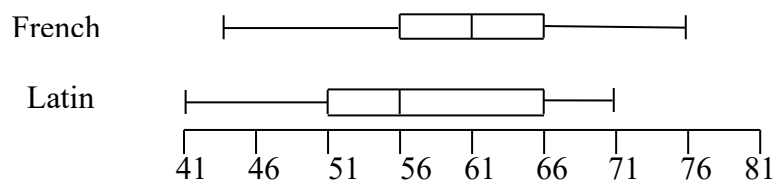
Given $y = 4 \cos\left(2x - \frac{\pi}{4}\right)$ the period and amplitude is

- (A) amplitude = 2 and period = π
- (B) amplitude = 4 and period = π
- (C) amplitude = 4 and period = 2π
- (D) amplitude = 4 and period = 3π

Question 5

At a High School, 20 students study HSC French and 20 students study HSC Latin.

Their Trial HSC results in these subjects are shown in this double box-and-whisker plot:



Which one of the following statements **must** be true?

- (A) Ten students scored less than 51 in Latin.
- (B) 75% of French students scored more than 61 in the test
- (C) In Latin, more students scored 66 or more than between 41 and 66.
- (D) The median score for French is 5 more than the median score for Latin

Question 6

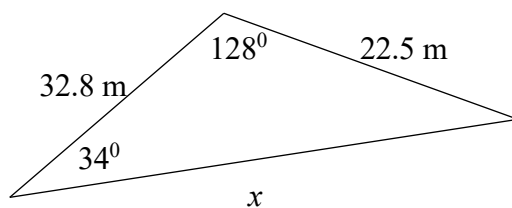
If $y = 3^{4x-5}$, then $\frac{dy}{dx} =$

- (A) $4(3^{4x-5})\ln 3$
- (B) $3(3^{4x-5})\ln 3$
- (C) $4(3^{4x-5})\ln 4$
- (D) $4(3^{4x-4})\ln 3$

Question 7

The correct value for x is:

Not to scale



- (A) $x = 22.5 \sin 128^\circ \div \sin 34^\circ$
- (B) $x = 22.5 \sin 34^\circ \div \sin 128^\circ$
- (C) $x = 22.5^2 + 32.8^2 - 22.5 \times 32.8 \times \cos 128^\circ$
- (D) $x = 22.5^2 + 32.8^2 - 22.5 \times 32.8 \times \cos 34^\circ$

Question 8

$$\int \cos 2x \, dx =$$

- (A) $\frac{1}{2} \sin 2x + c$
- (B) $\frac{-1}{2} \sin 2x + c$
- (C) $2 \sin 2x + c$
- (D) $\sin 4x + c$

Question 9

If $y = \log_2 4x$ then $\frac{dy}{dx} =$

(A) $\frac{\log_e 4x}{\log_e 8}$

(B) $\frac{1}{x \log_e 2}$

(C) $\frac{\log_e 2x}{\log_e 2}$

(D) $\frac{\log_e 4}{\log_e 2x}$

Question 10

Consider the geometric series $1 + (6 - \sqrt{a}) + (6 - \sqrt{a})^2 + (6 - \sqrt{a})^3 + \dots$

If the above series is to have a limiting sum, which of the following statements is correct.

(A) $9 < a < 64$ and $a \neq 36$

(B) $36 < a < 49$

(C) $25 < a < 49$ and $a \neq 36$

(D) $9 < a < 81$

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Section II Answer Booklet 1

Centre Number

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Student Number

Section II

90 marks

Attempt Questions 11 to 35

Allow about 2 hour and 45 minutes for this section

Booklet I - Attempt Questions 11 – 19 (28 marks)

Booklet 2 - Attempt Questions 20 – 35 (62 marks)

Instructions

1. Write your Centre Number and Student Number at the top of this page
 2. Answer the questions in the spaces provided. Sufficient spaces are provided for typical responses.
 3. Your responses should include mathematical reasoning and/or calculations.
 4. Extra writing space is provided at the back of this booklet. If you use this space, clearly indicate which question you are answering.
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Question 11 (2 marks)

Two particles A and B move along a straight line so that their displacements, in centimetres, from the origin at time t seconds are given by

$$X_A = 4t^2 - t^3 \quad \text{and} \quad X_B = 4t - 6$$

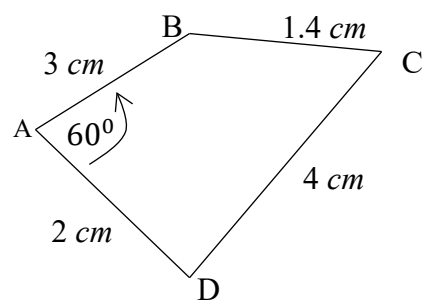
Which particle is moving faster when $t = 1$.

Question 12 (2 marks)

Between what values of x is there a solution to $\frac{x}{3} = 1 - 2\cos x$

Question 13 (5 Marks)

The shape of a new logo is shown below.



Find the area of the logo. Answer to one decimal place.

Question 14 (4 marks)

A survey was conducted on the heights of 1000 adults and the results were found to form a normal distribution. The mean height was 1.74 m and the standard deviation was 9 cm.

- (a) Approximately how many adults are between 1.65 m and 1.92 m. 2

- (b) Approximately how many adults will have a height less than 1.56 m? 2

Question 15 (2 marks)

A football coach is studying the relationship between the weight in kilograms of players and their speed over 150 metres, in metres per second. The data from ten players are shown in the table below.

weight	84.5	83.3	58.7	74.1	70.7	72.3	75.8	71.8	63.2	65.9
speed	21.9	20.5	20.8	19.9	20.6	18.8	22.8	20.6	20.0	21.8

Find Pearson's correlation coefficient of this data, and state any conclusions.

Question 16 (5 marks)

Which option would give the better result at the end of 2 years?

Option A: an annuity of \$1300 a quarter at 4% p.a., compounded quarterly.

Option B: an annuity of \$1400 every six months at 8% p.a. compounded half yearly.

Option C: \$1 500 invested at a simple interest rate of 10% p.a.

Question 17 (2 marks)

Data for life expectancy for males in a particular country are given in the table.

Age (x years)	Life expectancy for males (y years)
0	80.1
5	75.6
10	71.2
15	66.3
20	61.7
25	56.9
30	51.4
35	46.8
40	41.2
45	36.7

- (a) Find the equation of the least-squares regression line in the form $y = mx + c$, where m and c are correct to two decimal places.

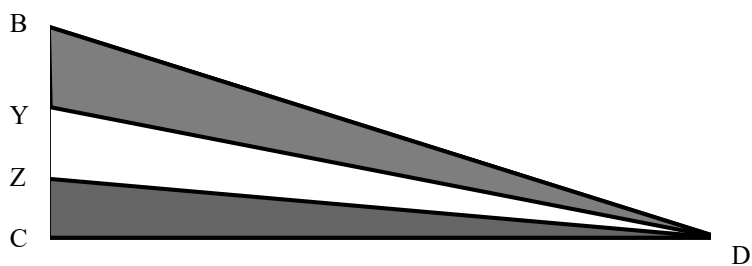
1

- (b) Doug is a male aged 42. Use the least-squares regression equation to calculate his life expectancy.

1

Question 18 (4 marks)

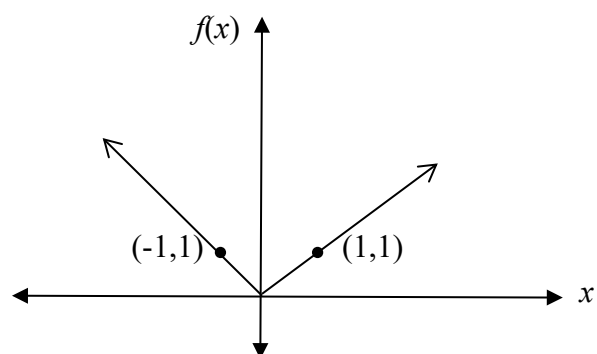
A flag shown is in three sections. The triangle BCD is right angled with $CD = 140$ cm. It is given $CZ = BY = YZ = 17$ cm and angle $YZD = 95^\circ$



What percentage of the area of the banner is the triangle YZD .

Question 19 (2 marks)

The curve $f(x) = |x|$ is shown below



On the same diagram sketch $y = 2[f(x) + 2]$

End of Question 19

Questions 11 to 19 are worth 28 marks

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Centre Number

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Student Number

Section II

Booklet 2 - Attempt Questions 20 – 35 (62 marks)

Instructions

Write your Centre Number and Student Number at the top of this page

Answer the questions in the spaces provided. Sufficient spaces are provided for typical responses.

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Question 20 (4 marks)

Let X be a continuous random variable with probability density function.

$$f(x) = \begin{cases} \frac{ax}{108}, & 28 \leq x \leq 30 \\ 0, & \text{for all other values of } x \end{cases}$$

(a) Find the value of a .

2

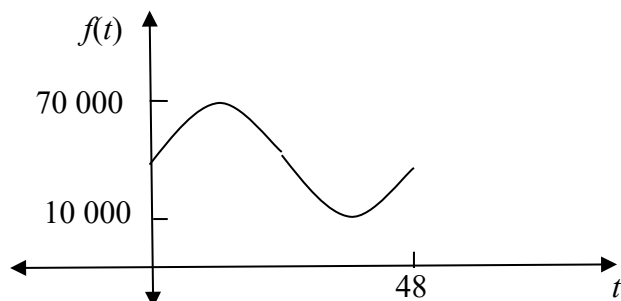
(b) Evaluate $P(28.5 \leq x \leq 29.5)$

2

Question 21 (2 marks)

The function $f(t) = a \sin\left(\frac{\pi}{24}t\right) + b$ is drawn below

The maximum value of $f(t)$ is 70 000 when $t=12$. The minimum of $f(t)$ is 10 000 when $t=36$. ($0 \leq t \leq 48$)



What are the values of a and b ?

Question 22 (3 marks)

A study was conducted to investigate the effect of beer consumption on sleep. In this study, the number of cans of beer consumed and the amount of sleep (in hours) were recorded for a group of adults.

	Beer (cans) (x)	Sleep (hours) (y)
Mean	2.32	6.9
Standard deviation	1.45	1.04
Correlation coefficient (r)	-0.699	

- (a) The equation of the least-squares line of best fit is $y = mx + c$ where

$$m = r \times \frac{\text{standard deviation of } y\text{-scores}}{\text{standard deviation of } x\text{-scores}}$$

$$c = \text{mean of } y \text{ scores} - m \times \text{mean of } x \text{ scores} = \bar{y} - m\bar{x}$$

Find the value of m

2

- (b) For each additional can of beer consumed, calculate how much the amount of sleep changes.

1

Question 23 (6 marks)

The gradient function of a curve is given by $f'(x) = (3x - 4)(x - 4)$ and the curve $y = f(x)$ passes through $(1, 9)$.

(a) Find the equation of the curve $y = f(x)$.

2

(b) Find any stationary points and determine their nature.

2

(c) Sketch the curve $y=f(x)$ clearly labelling turning points.

2

Question 24 (2 marks)

The function $f(x) = 4\tan^3 x$ is given.

If $f'(x) = 12 \tan^2 x (1 + \tan^m x)$, find the value of m .

End of Question 24

Questions 11 to 24 are worth 45 marks

Question 25 (3 marks)

Jill has five pairs of stockings in a box, each pair a different colour.
She selects one stocking at a time and at random from the box. Each time the stocking is not replaced back into the box.

- (a) Find the probability when she chooses her second stocking it will be a matching pair. **1**

- (b) Find the probability that she does not have a matching pair after selecting the third stocking. **2**

Question 26 (7 marks)

Kate inherits \$100 000 and invests it in an account earning interest at a rate of 0.6% per month. Each month, immediately after the interest has been paid, Kate withdraws \$1 200.

The amount in the account immediately after the n th withdrawal can be determined using the recurrence relation

$$A_n = A_{n-1}(1.006) - 1200,$$

Where $n = 1, 2, 3, \dots$ and $A_0 = 100\,000$

- (a) Use the above recurrence relation to find the amount of money in the account immediately after the third withdrawal. **2**

- (b) Calculate the amount of interest earned in the first three months. 2

- (c) Calculate the amount of money in the account immediately after the 75th withdrawal. 3

This image shows a blank sheet of white paper with horizontal ruling lines. The lines are evenly spaced and extend across the width of the page. There are no margins, text, or other markings on the paper.

Question 27 (4 marks)

(a) Differentiate $y = xe^{5x}$

2

(b) Hence, find the value of $\int_0^4 e^{5x} (5 + 25x) dx$

2

Question 28 (2 marks)

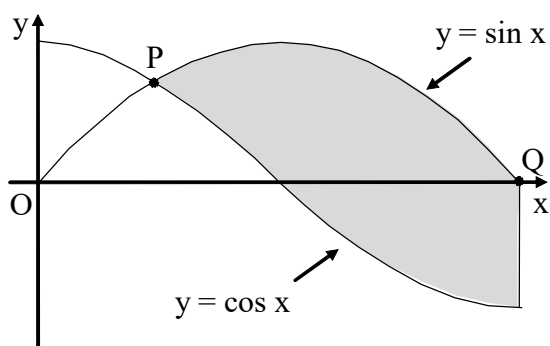
A pool is being drained and the number of litres of water, L , in the pool at time t minutes is given by the equation

$$L = 280(80 - t)^2$$

At what rate is the water draining out of the pool when $t = 10$ minutes.

Question 29 (4 marks)

The graph below shows the curves $y = \sin x$ and $y = \cos x$



(a) Find the x coordinates of the points P and Q.

1

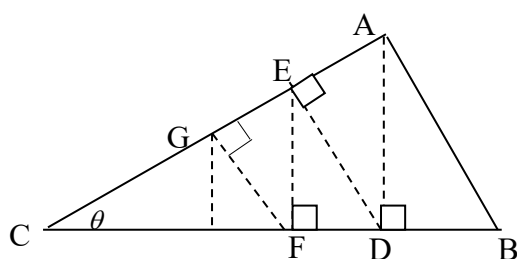
(b) Find the area of the shaded region.

3

Question 30 (5 marks)

In the triangle ABC , $\angle ADE = \angle ACB = \theta$ where $0 < \theta < \frac{\pi}{2}$ and AC is of length d .

A fly starts at A , flies directly to the line CB , i.e. to the point D . It then flies directly to the line CA , i.e. to the point E . It then flies directly to the line CB and so on until it ultimately reaches C .



- (a) Show that the distance travelled by the fly when it reaches the point E is $d \sin \theta (1 + \cos \theta)$

2

Question 31 (2 marks)

Given the function $f(x) = x^2 + 4$ and $g(x) = \sqrt{x - 8}$, sketch $y = f(g(x))$ over its natural domain.

Question 32 (4 marks)

A continuous random variable X has a probability density function f given by

$$f(x) = \begin{cases} Ax + B, & 1 \leq x \leq 6 \\ 0, & \text{otherwise} \end{cases}$$

where A and B are constants.

The median of X is 3

Find the values of A and B

Question 33 (4 marks)

Evaluate $\int_1^4 2x^2 (x^3 + 6)^3 dx$

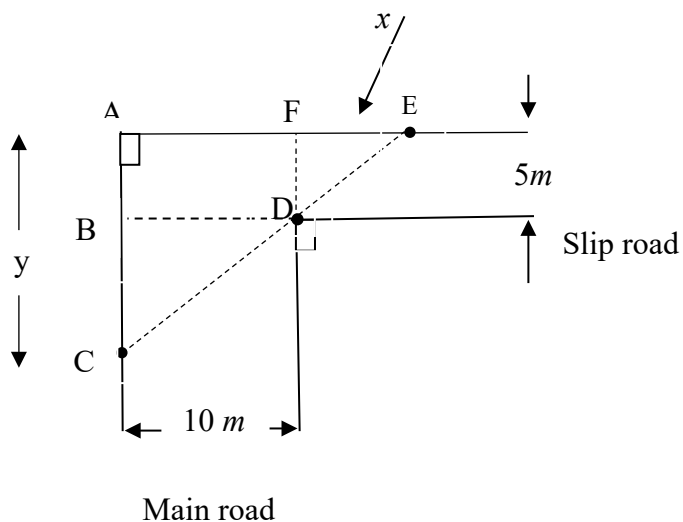
Question 34 (4 marks)

Henry goes to the gym on Fridays. He uses only the swimming pool or the rowing machines. On any given Friday:

$P(R|S) = \frac{3}{8}$ $P(S|R) = \frac{3}{10}$ $P(R \cup S) = \frac{7}{9}$ where R represents the event he uses the rowing machines and S represents the event he uses the swimming pool.

Calculate the values of $P(R)$ and $P(S)$

Question 35 (6 marks)



A slip road of width $5m$ is joined perpendicular to a main road of width $10m$ as shown in the diagram. Three street lamps are to be installed at positions C, D and E, where CDE form a straight line. C and E are variable points where $FE = x$ and $AC = y$

- (a) Show that $y = 5 + \frac{50}{x}$

1

4

[illegible]

- (c) By taking points either side of the stationary point, or otherwise, find the value of x for which the distances between the three lamps are a minimum. **1**

End of Exam

Section II Extra writing space

If you use this space, clearly indicate which question you are answering.

This image shows a full page of white paper with horizontal dashed lines, typical of primary-ruled notebook paper. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

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Measurement Length $l = \frac{\theta}{360} \times 2\pi r$ Area $A = \frac{\theta}{360} \times \pi r^2$ $A = \frac{h}{2}(a + b)$ Surface area $A = 2\pi r^2 + 2\pi rh$ $A = 4\pi r^2$ Volume $V = \frac{1}{3}Ah$ $V = \frac{4}{3}\pi r^3$	Financial Mathematics $A = P(1 + r)^n$ Sequences and series $T_n = a + (n - 1)d$ $S_n = \frac{n}{2}[2a + (n - 1)d] = \frac{n}{2}(a + l)$ $T_n = ar^{n-1}$ $S_n = \frac{a(1 - r^n)}{1 - r} = \frac{a(r^n - 1)}{r - 1}, r \neq 1$ $S = \frac{a}{1 - r}, r < 1$
Functions $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$ For $ax^3 + bx^2 + cx + d = 0$: $\alpha + \beta + \gamma = -\frac{b}{a}$ $\alpha\beta + \alpha\gamma + \beta\gamma = \frac{c}{a}$ $\text{and } \alpha\beta\gamma = -\frac{d}{a}$ Relations $(x - h)^2 + (y - k)^2 = r^2$	Logarithmic and Exponential Functions $\log_a a^x = x = a^{\log_a x}$ $\log_a x = \frac{\log_b x}{\log_b a}$ $a^x = e^{x \ln a}$

Trigonometric Functions

$$\sin A = \frac{\text{opp}}{\text{hyp}}, \cos A = \frac{\text{adj}}{\text{hyp}}, \tan A = \frac{\text{opp}}{\text{adj}}$$

$$\text{Area} = \frac{1}{2} ab \sin C$$

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

$$c^2 = a^2 + b^2 - 2ab \cos C$$

$$\cos C = \frac{a^2 + b^2 - c^2}{2ab}$$

$$l = r\theta$$

$$\text{Area} = \frac{1}{2} r^2 \theta$$

Trigonometric identities

$$\sec A = \frac{1}{\cos A}, \cos A \neq 0$$

$$\operatorname{cosec} A = \frac{1}{\sin A}, \sin A \neq 0$$

$$\cot A = \frac{\cos A}{\sin A}, \sin A \neq 0$$

$$\cos^2 x + \sin^2 x = 1$$

Compound angles

$$\sin(A+B) = \sin A \cos B + \cos A \sin B$$

$$\cos(A+B) = \cos A \cos B - \sin A \sin B$$

$$\tan(A+B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}$$

$$\text{If } t = \tan \frac{A}{2} \text{ then } \sin A = \frac{2t}{1+t^2}$$

$$\cos A = \frac{1-t^2}{1+t^2} \quad \tan A = \frac{2t}{1-t^2}$$

$$\cos A \cos B = \frac{1}{2} [\cos(A-B) + \cos(A+B)]$$

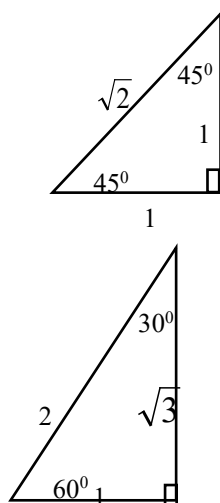
$$\sin A \sin B = \frac{1}{2} [\cos(A-B) - \cos(A+B)]$$

$$\sin A \cos B = \frac{1}{2} [\sin(A+B) + \sin(A-B)]$$

$$\cos A \sin B = \frac{1}{2} [\sin(A+B) - \sin(A-B)]$$

$$\sin^2 nx = \frac{1}{2} (1 - \cos 2nx)$$

$$\cos^2 nx = \frac{1}{2} (1 + \cos 2nx)$$



Statistical Analysis

$$z = \frac{x - \mu}{\sigma}$$

An outlier is a score

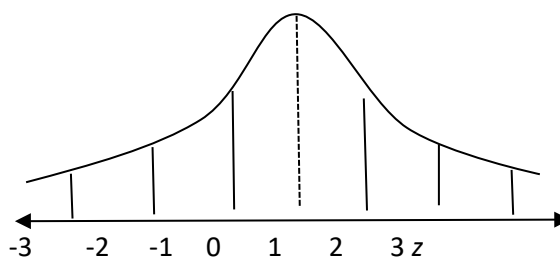
less than $Q_1 - 1.5 \times IQR$

or

more than

$$Q_3 + 1.5 \times IQR$$

Normal distribution



1. Approximately 68% of scores have z-scores between -1 and 1
2. Approximately 95% of scores have z-scores between -2 and 2
3. Approximately 99.7% of scores have z-scores between -3 and 3

$$E(X) = \mu$$

$$\operatorname{Var}(X) = E[(X - \mu)^2] = E(X^2) - \mu^2$$

Probability

$$P(A \cap B) = P(A)P(B)$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$P(A|B) = \frac{P(A \cap B)}{P(B)}, P(B) \neq 0$$

Continuous random variables

$$P(X \leq x) = \int_a^x f(x) dx$$

$$P(a < X < b) = \int_a^b f(x) dx$$

Binomial distribution

$$P(X = r) = {}^n C_r p^r (1-p)^{n-r}$$

$$X \sim \operatorname{Bin}(n, p)$$

$$\Rightarrow P(X = x)$$

$$= \binom{n}{x} p^x (1-p)^{n-x}, x = 0, 1, \dots, n$$

$$E(X) = np$$

$$\operatorname{Var}(X) = np(1-p)$$

Differential Calculus		Integral Calculus	
Function	Derivative		
$y = f(x)^n$	$\frac{dy}{dx} = nf'(x)[f(x)]^{n-1}$	$\int f'(x)[f(x)]^n dx = \frac{1}{n+1}[f(x)]^{n+1} + c$	where $n \neq -1$
$y = uv$	$\frac{dy}{dx} = u \frac{dv}{dx} + v \frac{du}{dx}$	$\int f'(x) \sin f(x) dx = -\cos f(x) + c$	
$y = g(u)$ where $u = f(x)$	$\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}$	$\int f'(x) \cos f(x) dx = \sin f(x) + c$	
$y = \frac{u}{v}$	$\frac{dy}{dx} = \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2}$	$\int f'(x) \sec^2 f(x) dx = \tan f(x) + c$	
$y = \sin f(x)$	$\frac{dy}{dx} = f'(x) \cos f(x)$	$\int f'(x) e^{f(x)} dx = e^{f(x)} + c$	
$y = \cos f(x)$	$\frac{dy}{dx} = -f'(x) \sin f(x)$	$\int \frac{f'(x)}{f(x)} dx = \ln f(x) + c$	
$y = \tan f(x)$	$\frac{dy}{dx} = f'(x) \sec^2 f(x)$	$\int f'(x) a^{f(x)} dx = \frac{a^{f(x)}}{\ln a} + c$	
$y = e^{f(x)}$	$\frac{dy}{dx} = f'(x) e^{f(x)}$	$\int \frac{f'(x)}{\sqrt{a^2 - [f(x)]^2}} dx = \sin^{-1} \frac{f(x)}{a} + c$	
$y = \ln f(x)$	$\frac{dy}{dx} = \frac{f'(x)}{f(x)}$	$\int \frac{f'(x)}{\sqrt{a^2 + [f(x)]^2}} dx = \frac{1}{a} \tan^{-1} \frac{f(x)}{a} + c$	
$y = a^{f(x)}$	$\frac{dy}{dx} = (\ln a) f'(x) a^{f(x)}$	$\int u \frac{dv}{dx} dx = uv - \int v \frac{du}{dx} dx$	
$y = \log_a f(x)$	$\frac{dy}{dx} = \frac{f'(x)}{(\ln a) f(x)}$	$\int_a^b f(x) dx \approx \frac{b-a}{2n} \{f(a) + f(b) + 2[f(x_1) + \dots + f(x_{n-1})]\}$	where $a = x_0$ and $b = x_n$
$y = \sin^{-1} f(x)$	$\frac{dy}{dx} = \frac{f'(x)}{\sqrt{1 - [f(x)]^2}}$		
$y = \cos^{-1} f(x)$	$\frac{dy}{dx} = -\frac{f'(x)}{\sqrt{1 - [f(x)]^2}}$		
$y = \tan^{-1} f(x)$	$\frac{dy}{dx} = \frac{f'(x)}{1 + [f(x)]^2}$		

Combinations

$${}^n P_r = \frac{n!}{(n-r)!}$$

$$\binom{n}{r} = {}^n C_r = \frac{n!}{r!(n-r)!}$$

$$(x+a)^n = x^n + \binom{n}{1} x^{n-1} a + \dots + \binom{n}{r} x^{n-r} a^r + \dots + a^n$$

Vectors

$$|U| = |xi+yj| = \sqrt{x^2 + y^2}$$

~ ~ ~

$$u \bullet v = |u| |v| \cos \theta = x_1 x_2 + y_1 y_2$$

~ ~ ~ ~

Where $u = x_1 i + y_1 j$

~ ~

and $v = x_2 i + y_2 j$

~ ~ ~

$$r = a + \lambda b$$

~ ~

Complex Numbers

$$z = a + ib = r(\cos \theta + i \sin \theta)$$

$$= r e^{i\theta}$$

$$[r(\cos \theta + i \sin \theta)]^n = r^n (\cos n\theta + i \sin n\theta)$$

$$= r^n e^{in\theta}$$

Mechanics

$$\frac{d^2 x}{dt^2} = \frac{dv}{dt} = v \frac{dv}{dx} = \frac{d}{dx} \left(\frac{1}{2} v^2 \right)$$

$$x = a \cos(nt + \alpha) + c$$

$$x = a \sin(nt + \alpha) + c$$

$$\ddot{x} = -n^2(x - c)$$

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YEAR 12 Trial HSC EXAMINATION

MATHEMATICS Advanced-MULTIPLE CHOICE ANSWER SHEET

Sample $2 + 4 = ?$ (A) 2 (B) 6 (C) 8 (D) 9

 (A)  (B)  (C)  (D) 

If you think you have made a mistake, put a cross through the incorrect answer and fill in the new answer.

(A) (B) (C) (D)

If you change your mind and have crossed out what you consider to be the correct answer, then indicate this by writing the word *correct* and drawing an arrow as follows *correct*

(A)  (B)  (C)  (D) 

correct

- 1 (A) ☐ (B) ☐ (C) ☐ (D) ☐
- 2 (A) ☐ (B) ☐ (C) ☐ (D) ☐
- 3 (A) ☐ (B) ☐ (C) ☐ (D) ☐
- 4 (A) ☐ (B) ☐ (C) ☐ (D) ☐
- 5 (A) ☐ (B) ☐ (C) ☐ (D) ☐
- 6 (A) ☐ (B) ☐ (C) ☐ (D) ☐
- 7 (A) ☐ (B) ☐ (C) ☐ (D) ☐
- 8 (A) ☐ (B) ☐ (C) ☐ (D) ☐
- 9 (A) ☐ (B) ☐ (C) ☐ (D) ☐
- 10 (A) ☐ (B) ☐ (C) ☐ (D) ☐

2022

Trial HSC MATHEMATICS ADVANCED

Marking Guidelines

Section I

Multiple-choice Answer key

1	D
2	D
3	D
4	B
5	D
6	A
7	A
8	A
9	B
10	C

We have endeavored to ensure that the solutions are free of errors and follow the spirit of the syllabus in the methods used to solve the problems. However, individual teachers may opt for alternate solutions and/or may choose a different marking system.

Question 11

Criteria	Marks
• Provides the correct answer	2
• Provides some relevant information	1

$$X_A = 4t^2 - t^3$$

$$\dot{X}_A = 8t - 3t^2$$

when $t = 1$

$$\dot{X}_A = 5 \text{ cm / sec}$$

$$X_B = 4t - 6$$

$$\dot{X}_B = 4$$

when $t = 1$

$$\dot{X}_B = 4 \text{ cm / sec}$$

$\therefore \dot{X}_A$ is faster at $t = 1$

Question 12

Criteria	Marks
• Provides the correct answer	2
• Provides some relevant information	1

$$\frac{x}{3} = 1 - 2\cos x$$

$$\therefore 2\cos x = 1 - \frac{x}{3}$$

$$\text{Since } -1 \leq \cos x \leq 1$$

$$\therefore -2 \leq 2\cos x \leq 2$$

$$\therefore -2 \leq 1 - \frac{x}{3} \leq 2$$

$$-3 \leq -\frac{x}{3} \leq 1$$

$$9 \geq x \geq -3$$

$$-3 \leq x \leq 9$$

$$\text{i.e. } [-3, 9]$$

Question 13

Criteria	Marks
• Provides correct answer	5
• Finds area BCD	4
• Uses cosine rule to find BD	3
• Correctly finds area of one triangle	2
• Provides some relevant information	1

Area ABD

$$A = \frac{1}{2} \times 3 \times 2 \times \sin 60^\circ = 2.60$$

Length BD

$$BD^2 = 3^2 + 2^2 - 2 \times 3 \times 2 \times \cos 60^\circ = 7$$

$$BD = 2.65$$

Let Angle BCD = θ

$$\cos \theta = \frac{1.4^2 + 4^2 - 2.65^2}{2 \times 1.4 \times 4} = 0.97656$$

$$\theta = 12.429^\circ$$

Area of BCD

$$A = \frac{1}{2} \times 1.4 \times 4 \times \sin 12.429^\circ = 0.6026$$

$$\text{Total area of ABCDA} = 2.6 + 0.6026 = 3.2 \text{ cm}^2$$

Question 14 (a)

Criteria	Marks
• Provides the correct answer	2
• Provides some relevant information	1

A score of 1.65m is one standard deviation below the mean. A score of 1.92m is two standard deviations above the mean.

Therefore between 1.65 and 1.92 = $34\% + 47.5\% = 81.5\%$

Therefore 81.5% of 1000 = 815

Question 14 (b)

Criteria	Marks
• Provides the correct answer	2
• Provides some relevant information	1

2.5% of adults will have a height below 1.56 m.

i.e. 2.5% of 1000 = 25

Question 15

Criteria	Marks
• Provides the correct answer	2
• Provides some relevant information	1

0.1992.

There is only a moderate positive correlation between the weight of the player and their speed over 150 metres.

Question 16

Criteria	Marks
• Provides the correct answer	5
• Both future values for option A and B are correct	4
• Future value for option A or B is correct	3
• Option C has correct answer	2
• Provides some relevant information	1

Option A:

$$\text{Interest rate} = (4/4)\% = 1\%$$

$$\text{Number of periods} = 2 \times 4 = 8$$

$$\text{Future value} = \$1300 \times (1 + 1/100)^8 = \$1\,407.71$$

Option B:

$$\text{Interest rate} = (8/2)\% = 4\%$$

$$\text{Number of periods} = 2 \times 2 = 4$$

$$\text{Future value} = \$1400 \times (1 + 4/100)^4 = \$1\,637.80$$

Option C:

$$\text{Simple interest} = 1500 \times 10/100 = \$150$$

$$\text{Future value} = 1500 + 150 = \$1650$$

Option B is the better result after 2 years.

Question 17 (a)

Criteria	Marks
• Provides the correct answer	1

$$y = -0.97x + 80.69$$
$$y = -0.97 \times 42 + 80.69 = 39.95$$

Question 17 (b)

Criteria	Marks
• Provides the correct answer	1

Approx. 40 years

Question 18

Criteria	Marks
• Provides correct answer	4
• Finds area of triangle ZCD and BCD	3
• Uses Pythagoras rule correctly	2
• Provides some relevant information	1

$$ZD^2 = 17^2 + 140^2 = 19889 \text{ thus } ZD = 141.028$$

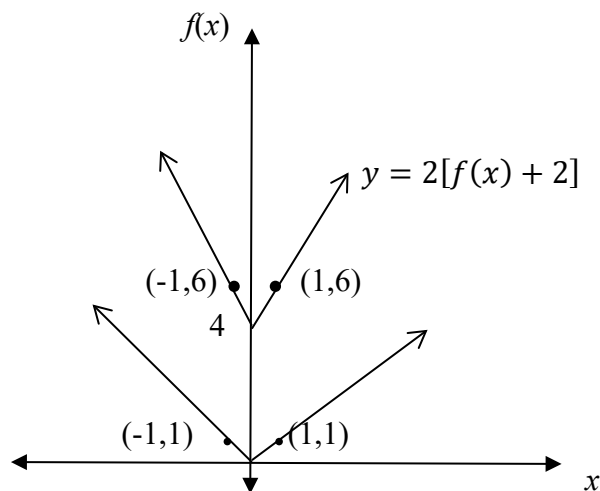
$$\text{Area of triangle YZD} = \frac{1}{2} \times 141.028 \times 17 \times \sin 95^\circ = 1194.176$$

$$\text{Area of triangle BCD} = \frac{1}{2} \times 140 \times 3 \times 17 = 3570$$

$$\text{Percentage} = \frac{1194.176}{3570} \times 100\% = 33.45 = 33.5\%$$

Question 19

Criteria	Marks
• Provides the correct answer	2
• Provides some relevant information	1



Question 20 (a)

Criteria	Marks
• Provides the correct answer	2
• Correct integration	1

$$\int_{28}^{30} \frac{ax}{108} dx = 1$$

$$\begin{aligned} \left[\frac{ax^2}{216} \right]_{28}^{30} &= 1 \\ \frac{900a}{216} - \frac{784a}{216} &= 1 \\ \frac{116a}{216} &= 1 \therefore a = \frac{54}{29} \end{aligned}$$

Question 20 (b)

Criteria	Marks
• Provides correct answer	2
• Provides correct integration	1

$$P(28.5 \leq x \leq 29.5) = \int_{28.5}^{29.5} \frac{x}{58} dx$$

$$\left[\frac{x^2}{116} \right]_{28.5}^{29.5} = \frac{1}{2}$$

Question 21

Criteria	Marks
• Provides the correct answer	2
• Provides some relevant information	1

$$\text{amplitude, } a = \frac{70\,000 - 10\,000}{2} = 30\,000$$

$$\text{vertical intercept } f(12) = 70\,000 = 30\,000 \sin\left(\frac{\pi}{24} \times 12\right) + b$$

$$b = 40\,000$$

Question 22 (a)

Criteria	Marks
• Provides the correct answer	2
• Provides some relevant information	1

$$m = r \times \frac{\text{standard deviation of } y - \text{scores}}{\text{standard deviation of } x - \text{scores}}$$

$$m = -0.699 \times \frac{1.04}{1.45} = -0.501..$$

Question 22 (b)

Criteria	Marks
• Provides the correct answer	1

The amount of sleep decreases by 0.5 h for each additional beer consumed.

Question 23 (a)

Criteria	Marks
• Provides the correct answer	2
• Provides some relevant information	1

$$f'(x) = 3x^2 - 16x + 16$$

$$\therefore f(x) = \int (3x^2 - 16x + 16) dx$$

$$= x^3 - 8x^2 + 16x + C$$

(1,9) is a point on $y = f(x)$

i.e. $9 = 1 - 8 + 16 + C$

$$\therefore C = 0$$

Equation of the curve is

$$y = x^3 - 8x^2 + 16x = x(x-4)^2$$

Question 23 (b)

Criteria	Marks
• Provides the correct answer	2
• Provides some relevant information	1

$$y = x^3 - 8x^2 + 16x$$

$$f'(x) = 3x^2 - 16x + 16 = (3x - 4)(x - 4)$$

At a stationary point $f'(x) = 0$

$$\therefore (3x - 4) = 0 \text{ or } x - 4 = 0$$

i.e $x = 4/3$ or 4

$$f''(x) = 6x - 16$$

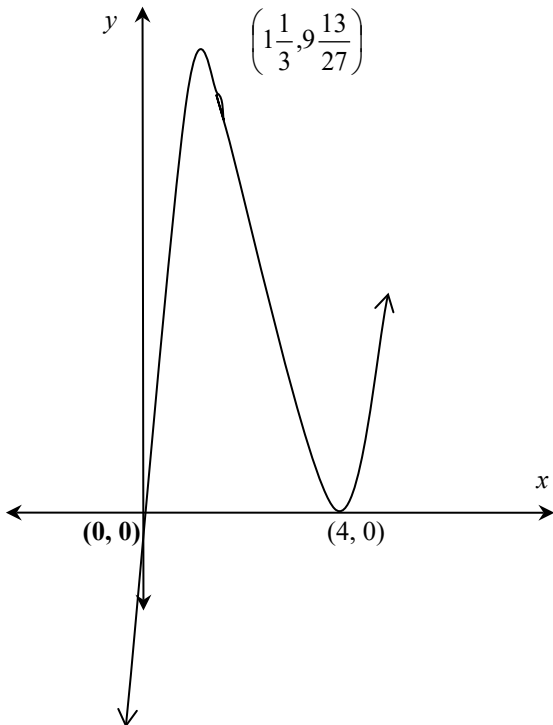
$$f''\left(\frac{4}{3}\right) < 0 \text{ and } f''(4) > 0$$

$\therefore$ At $x = 4/3$, $f(x)$ is maximum and
at $x = 4$, $f(x)$ is minimum

$\max \left(\frac{4}{3}, 9\frac{13}{27} \right)$ and $\min (4, 0)$

Question 23 (c)

Criteria	Marks
• Provides the correct answer	2
• Provides some relevant information	1



Question 24

Criteria	Marks
• Provides the correct answer	2
• Provides some relevant information	1

$$\begin{aligned}
 f(x) &= 4\tan^3 x \\
 f(x) &= 4(\tan x)^3 \\
 f'(x) &= 12\tan^2 x \sec^2 x \\
 &= 12\tan^2 x(1 + \tan^2 x) \\
 &= 12\tan^2 x + 12\tan^4 x \\
 &= 12\tan^2 x(1 + \tan^2 x) \therefore m = 2
 \end{aligned}$$

Question 25 (a)

Criteria	Marks
• Provides the correct answer	1

Let the stockings be G G R R O O B B P P

$$P(\text{second stocking will be a matching pair}) = 1 \times \frac{1}{9}$$

$$= \frac{1}{9} \quad \text{If G is selected first then probability of G second is } \frac{1}{9}$$

Question 25 (b)

Criteria	Marks
• Provides the correct answer	2
• Provides some relevant information	1

Answer:

$$P(\text{not having a matching pair after selecting the third stocking}) = \frac{10}{10} \times \frac{8}{9} \times \frac{6}{8} = \frac{2}{3}$$

Question 26 (a)

Criteria	Marks
• Provides the correct answer	2
• Provides some relevant information	1

$$A_0 = 100\,000$$

$$A_1 = A_0(1.006) - 1\,200 = 100\,000(1.006) - 1\,200 = 99\,400$$

$$A_2 = A_1(1.006) - 1\,200 = 99\,400(1.006) - 1\,200 = 98\,796.4$$

$$A_3 = A_2(1.006) - 1\,200 = 98\,796.4(1.006) - 1\,200 = \$98\,189.18$$

Question 26 (b)

Criteria	Marks
• Provides the correct answer	2
• Provides some relevant information	1

$$100\,000 - 1\,200 \times 3 + I = 98\,189.18$$

$$96\,400 + I = 98\,189.18 \therefore I = \$1\,789.18$$

Question 26 (c)

Criteria	Marks
• Provides the correct answer	3
• Makes significant progress	2
• Provides some relevant information	1

$$A_0 = 100\,000$$

$$A_1 = 100\,000(1.006) - 1\,200$$

$$A_2 = [100\,000(1.006) - 1\,200](1.006) - 1\,200$$

$$A_2 = 100\,000(1.006)^2 - 1\,200(1.006) - 1\,200$$

$$A_3 = [100\,000(1.006)^2 - 1\,200(1.006) - 1\,200](1.006) - 1\,200$$

$$A_3 = 100\,000(1.006)^3 - 1\,200(1.006)^2 - 1\,200(1.006) - 1\,200$$

$$A_3 = 100\,000(1.006)^3 - 1\,200(1 + 1.006 + 1.006^2)$$

$$A_n = 100\,000(1.006)^n - 1\,200(1 + 1.006 + 1.006^2 + \dots + 1.006^{n-1})$$

$$A_n = 100\,000(1.006)^n - 1\,200 \times \frac{1.006^n - 1}{0.006}$$

$$A_{75} = 100\,000(1.006)^{75} - 1\,200 \times \frac{1.006^{75} - 1}{0.006} = \$433\,799.52$$

Question 27(a)

Criteria	Marks
• Provides the correct answer	2
• Provides some information leading to correct answer	1

$$y = xe^{5x}$$

$$u = x \therefore \frac{du}{dx} = 1$$

$$v = e^{5x} \therefore \frac{dv}{dx} = 5e^{5x}$$

$$\begin{aligned}\frac{dy}{dx} &= 1 \times e^{5x} + 5xe^{5x} \\ &= e^{5x}(1 + 5x)\end{aligned}$$

Question 27(b)

Criteria	Marks
• Provides the correct answer	2
• Provides some information leading to correct answer	1

$$\begin{aligned}&\int_0^4 e^{5x} (5 + 25x) dx \\ &= 5 \int_0^4 e^{5x} (1 + 5x) dx \\ &= 5[xe^{5x}]_0^4 \\ &= 5(4e^{20} - 0) \\ &= 20e^{20}\end{aligned}$$

Question 28

Criteria	Marks
• Provides the correct answer	2
• Provides some information leading to correct answer	1

$$\begin{aligned}L &= 280(80 - t)^2 \\ \frac{dL}{dt} &= -560(80 - t) \\ \text{at } t = 10 \therefore L &= -39200\end{aligned}$$

$\therefore$ draining out at 39200 litres per minute

Question 29 (a)

Criteria	Marks
• Provides the correct answer	1

$$\sin x = \cos x$$

$$\tan x = 1 \therefore x = \frac{\pi}{4} \therefore \text{the } x \text{ coordinate of P is } \frac{\pi}{4}$$

$$\text{the } x \text{ coordinate of Q is } \pi$$

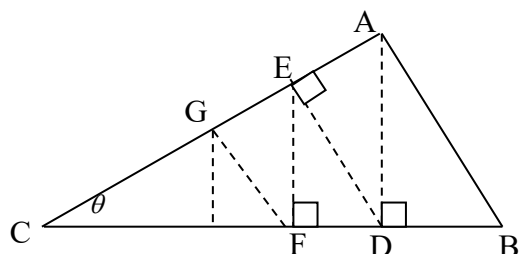
Question 29 (b)

Criteria	Marks
• Provides the correct answer	3
• Makes significant progress	2
• Provides some relevant information	1

$$\begin{aligned}
 A &= \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \sin x - \cos x \, dx \\
 &= [-\cos x - \sin x]_{\frac{\pi}{4}}^{\frac{\pi}{2}} \\
 &= \left[\left(-\cos \frac{\pi}{2} - \sin \frac{\pi}{2} \right) - \left(-\cos \frac{\pi}{4} - \sin \frac{\pi}{4} \right) \right] \\
 &= (0 - 1) - \left(-\frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2} \right) = \sqrt{2} - 1 \text{ u}^2
 \end{aligned}$$

Question 30 (a)

Criteria	Marks
• Provides the correct answer	2
• Makes significant progress	1



Let D, E, F, G be the points the fly touches the inside of the triangle. From $\triangle ADC$

$$AD = AC \sin \theta$$

From $\triangle ADE$

$$DE = AD \cos \theta = d \sin \theta \cos \theta$$

$$\therefore \text{distance} = d \sin \theta + d \sin \theta \cos \theta$$

$$= d \sin \theta (1 + \cos \theta)$$

Question 30 (b)

Criteria	Marks
• Provides the correct answer	3
• Makes significant progress	2
• Provides some relevant information	1

Let D, E, F, G, be the points where the fly touches inside of the triangle.

From $\triangle ADC$, $AD = AC \sin \theta$

$$= d \sin \theta$$

From $\triangle ADE$, $DE = AD \cos \theta = d \sin \theta \cos \theta$

From $\triangle DEF$, $EF = DE \cos \theta = d \sin \theta \cos^2 \theta$

From $\triangle EFG$, $FG = EF \cos \theta = d \sin \theta \cos^3 \theta$

The total distance travelled is given by

$$s = d \sin \theta + d \sin \theta \cos \theta + d \sin \theta \cos^2 \theta + \dots$$

This is an infinite geometric series where

$$a = d \sin \theta \quad \text{and} \quad r = \cos \theta \quad \text{and} \quad |r| < 1$$

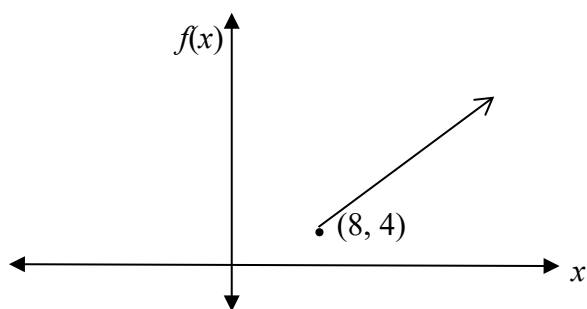
$$\therefore s = \frac{d \sin \theta}{1 - \cos \theta}$$

Question 31

Criteria	Marks
• Provides the correct answer	2
• Makes reasonable progress	1

$$\begin{aligned}f(g(x)) &= (\sqrt{x-8})^2 + 4 \\&= x - 8 + 4 \\&= x - 4\end{aligned}$$

The domain of $g(x)$ is $x \geq 8$. So $f(g(x))$ only exists for $x \geq 8$



Question 32

Criteria	Marks
• Provides the correct answer	4
• Evaluates integrals to produce two equations in A and B	3
• Makes reasonable progress	2
• Writes $\int_1^6 Ax + Bdx = 1$ or $\int_1^3 Ax + Bdx = \frac{1}{2}$ or equivalent merit	1

$$\begin{aligned}
 \int_1^6 Ax + Bdx &= 1 \\
 \left[\frac{Ax^2}{2} + Bx \right]_1^6 &= 1 \\
 18A + 6B - \left(\frac{A}{2} + B \right) &= 1 \\
 \frac{35A}{2} + 5B &= 1 \\
 35A + 10B &= 2 \dots\dots A \\
 \int_1^3 Ax + Bdx &= \frac{1}{2} \\
 \left[\frac{Ax^2}{2} + Bx \right]_1^3 &= \frac{1}{2} \\
 \frac{9}{2}A + 3B - \left(\frac{A}{2} + B \right) &= \frac{1}{2} \\
 4A + 2B &= \frac{1}{2} \\
 8A + 4B &= 1 \dots\dots B \\
 B \times \frac{10}{4} & \\
 20A + 10B &= \frac{10}{4} \dots\dots C \\
 A - C & \\
 A = \frac{-1}{30} \text{ and } B &= \frac{19}{60}
 \end{aligned}$$

Question 33

Criteria	Marks
• Provides the correct answer	4
• Majority of solution is correct	3
• Makes reasonable progress	2
• Shows some relevant information	1

$$\begin{aligned}
 & \int_1^4 2x^2(x^3 + 6)^3 dx \\
 &= \frac{2}{3} \int_1^4 3x^2(x^3 + 6)^3 dx \\
 &= \frac{2}{3} \times \frac{1}{4} [(x^3 + 6)^4]_1^4 \\
 &= \frac{1}{6} [(x^3 + 6)^4]_1^4 \\
 &= \frac{1}{6} (70 - 7) = \frac{63}{6} = 10\frac{1}{2}
 \end{aligned}$$

Question 34

Criteria	Marks
• Provides the correct answer	4
• Majority of solution is correct	3
• Makes reasonable progress	2
• Shows some relevant information	1

$$P(R|S) = \frac{P(R \cap S)}{P(S)} = \frac{3}{8}$$

$$\therefore P(R \cap S) = \frac{3}{8}P(S)$$

$$P(S|R) = \frac{P(R \cap S)}{P(R)} = \frac{3}{10}$$

$$\therefore P(R \cap S) = \frac{3}{10}P(R)$$

$$\frac{3}{8}P(S) = \frac{3}{10}P(R)$$

$$\therefore P(S) = \frac{4}{5}P(R)$$

$$P(R \cup S) = P(R) + P(S) - P(R \cap S)$$

$$\frac{7}{9} = P(R) + \frac{4}{5}P(R) - \frac{3}{10}P(R)$$

$$\therefore P(R) = \frac{14}{27}$$

$$\therefore P(S) = \frac{56}{135}$$

Question 35(a)

Criteria	Marks
• Provides the correct answer	1

$$\frac{FE}{FD} = \frac{AE}{AC}$$

$$\frac{x}{5} = \frac{x+10}{y}$$

$$y = \frac{5x+50}{x} = 5 + \frac{50}{x}$$

Question 35(b)

Criteria	Marks
• Provides the correct answer	4
• Majority of solution is correct	3
• Makes reasonable progress	2
• Shows some relevant information	1

$$p = \sqrt{x^2 + 25}$$

$$\frac{dp}{dx} = \frac{1}{2}(x^2 + 25)^{-\frac{1}{2}} \times 2x$$

$$= \frac{x}{\sqrt{x^2 + 25}}$$

$$Q = \sqrt{100 + (y - 5)^2}$$

$$= \sqrt{100 + \left(5 + \frac{50}{x} - 5\right)^2}$$

$$= \sqrt{100 + \frac{2500}{x^2}}$$

$$\frac{dQ}{dx} = \frac{1}{2}\left(100 + \frac{2500}{x^2}\right)^{-\frac{1}{2}} \times \frac{-5000}{x^3}$$

$$= \frac{\frac{-2500}{x^3}}{\sqrt{100 + \frac{2500}{x^2}}}$$

$$= \frac{\frac{-2500}{x^3}}{\frac{\sqrt{100x^2 + 2500}}{x}}$$

$$\begin{aligned}
 &= \frac{\frac{-2500}{x^3}}{10\sqrt{x^2+25}} \\
 &= \frac{\frac{x}{-250}}{x^2\sqrt{x^2+25}} \\
 \frac{dL}{dx} &= \frac{dP}{dx} + \frac{dQ}{dx} \\
 &= \frac{x}{\sqrt{x^2+25}} - \frac{250}{x^2\sqrt{x^2+25}} \\
 \frac{dL}{dx} &= \frac{x^3 - 250}{x^2\sqrt{x^2+25}}
 \end{aligned}$$

Question 35 (c)

Criteria	Marks
• Provides the correct answer	1

$$\begin{aligned}
 \frac{dL}{dx} &= \frac{x^3 - 250}{x^2\sqrt{x^2+25}} = 0 \\
 x^3 - 250 &= 0 \\
 x &= \sqrt[3]{250} \\
 x &= 6.2 \quad \sqrt[3]{250} \quad 6.3 \\
 \frac{dL}{dx} &= -0.0381 \quad 0 \quad 0.000147
 \end{aligned}$$

$\therefore$ minimum distance is when $x = \sqrt[3]{250}$

Examination Mapping Grid

Section 1

Question	Marks	Content	Syllabus outcomes	Targeted performance bands
1	1	MA-S2 Descriptive Statistics and Bivariate Data Analysis	MA 12-8	2-3
2	1	MA-S2 Descriptive Statistics and Bivariate Data Analysis	MA 12-8	2-3
3	1	MA-S1 Probability and Discrete Probability Distributions	MA12-8	2-3
4	1	MA-T3 Trigonometric Functions and Graphs	MA 12-5	2-3
5	1	MA-S2 Descriptive Statistics and Bivariate Data Analysis	MA 12-8	2-3
6	1	MA- C3 Applications of Differentiation	MA 12-3	2-3
7	1	MA-T1 Trigonometry and Measure of Angles	MA 11-3	2-3
8	1	MA- C3 Applications of Differentiation	MA 12-3	2-3
9	1	MA- C3 Applications of Differentiation	MA 12-3	2-3
10	1	MA-M1 Modelling Financial Situations	MA12-2	3-4

Section 2

11	2	MA- C3 Applications of Differentiation	MA 12-3	2-3
12	2	MA-T3 Trigonometric Functions and Graphs	MA 12-5	2-3
13	5	MA-T1 Trigonometry and Measure of Angles	MA 11-3	2-3
14 (a)	2	MA-S2 Descriptive Statistics and Bivariate Data Analysis	MA 12-8	2-3
14 (b)	2	MA-S2 Descriptive Statistics and Bivariate Data Analysis	MA 12-8	2-3
15	2	MA-S2 Descriptive Statistics and Bivariate Data Analysis	MA 12-8	2-3
16	5	MA-M1 Modelling Financial Situations	MA12-2	3-4
17 (a)	1	MA-S2 Descriptive Statistics and Bivariate Data Analysis	MA 12-8	2-3
17 (b)	1	MA-S2 Descriptive Statistics and Bivariate Data Analysis	MA 12-8	2-3
18	4	MA-T1 Trigonometry and Measure of Angles	MA 11-3	2-3
19	2	MA-F2 Graphing Techniques	MA12-1	3-4
20 (a)	2	MA-S1 Probability and Discrete Probability Distributions	MA12-8	2-3
20 (b)	2	MA-S1 Probability and Discrete Probability Distributions	MA12-8	2-3
21	2	MA- C3 Applications of Differentiation	MA 12-3	2-3
22 (a)	2	MA-S2 Descriptive Statistics and Bivariate Data Analysis	MA 12-8	2-3
22 (b)	1	MA-S2 Descriptive Statistics and Bivariate Data Analysis	MA 12-8	2-3
23 (a)	2	MA- C3 Applications of Differentiation	MA 12-3	3-4
23 (b)	2	MA- C3 Applications of Differentiation	MA 12-3	3-4
23(c)	2	MA- C3 Applications of Differentiation	MA 12-3	3-4
24	2	MA- C3 Applications of Differentiation	MA 12-3	3-4
25 (a)	1	MA-C4 Integral Calculus	MA 12-7	4-5
25 (b)	2	MA-T2 Trigonometric Functions and Identities	MA 11-4	3-4

26 (a)	2	MA-M1 Modelling Financial Situations	MA 12-2	3-4
26 (b)	2	MA-M1 Modelling Financial Situations	MA 12-2	3-4
26 (c)	3	MA-M1 Modelling Financial Situations	MA12-2	3-4
27 (a)	2	MA- C3 Applications of Differentiation	MA 12-3	3-4
27 (b)	2	MA-C4 Integral Calculus	MA 12-7	4-5
28	2	MA- C3 Applications of Differentiation	MA 12-3	3-4
29 (a)	1	MA-C4 Integral Calculus	MA 12-7	4-5
29 (b)	3	MA-C4 Integral Calculus	MA 12-7	4-5
30 (a)	2	MA-M1 Modelling Financial Situations	MA12-2	4-5
30 (b)	3	MA-M1 Modelling Financial Situations	MA12-2	3-4
31	2	MA-F2 Graphing Techniques	MA12-1	3-4
32	4	MA-S1 Probability and Discrete Probability Distributions	MA12-8	4-5
33	4	MA-C4 Integral Calculus	MA 12-7	4-5
34	4	MA-S1 Probability and Discrete Probability Distributions	MA12-8	3-4
35 (a)	1	MA- C3 Applications of Differentiation	MA 12-3	3-4
35 (b)	4	MA- C3 Applications of Differentiation	MA 12-3	5-6
35 (c)	1	MA- C3 Applications of Differentiation	MA 12-3	4-5